

→ Maxwellove enačbe X → Kirchhoffova zakona

$$\sum_k I_k = 0$$

↓
v vozliščih

$$\sum_n U_n = 0$$

↓
po zaključeni zanki

→ linearna vezja (vzročnost, homogenost, aditivnost, časovna neodvisnost)

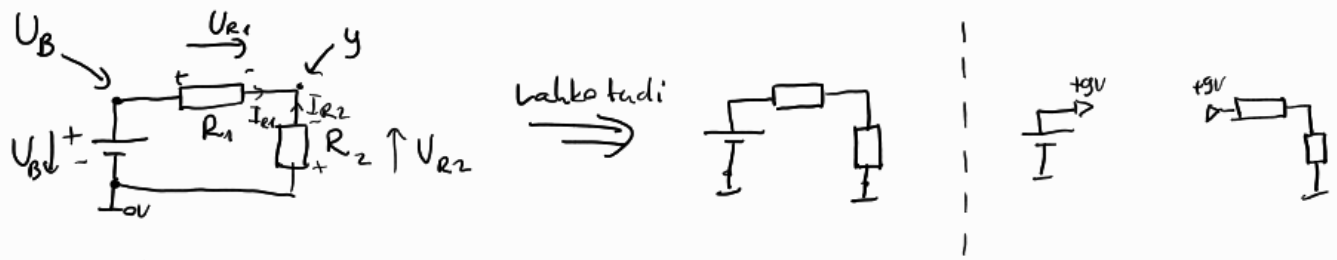
do odziva pride trenutek po vzbujanju; brez vzbujanja ni odziva (npr. izhod se ne bo spremenil če vhod ostane isti)

vzbujanje a
odziv b
⇒ vz. 2a
⇒ odz. 2b

če ob razl. časih vzbujamo vezje z istim signalom, dobimo isti odziv

UPORABA
- če se da signal razbit
- če ima vezje več vhodov

→ Ohmov zakon $R = \frac{U}{I}$



$$I_{R1} + I_{R2} = 0$$

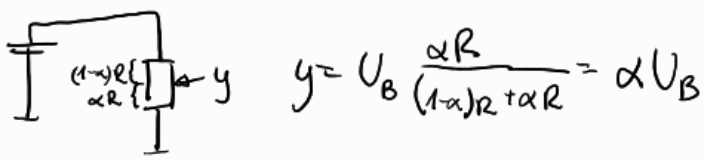
$$\frac{U_{R1}}{R_1} + \frac{U_{R2}}{R_2} = 0$$

$$\frac{U_B - y}{R_1} + \frac{0 - y}{R_2} = 0$$

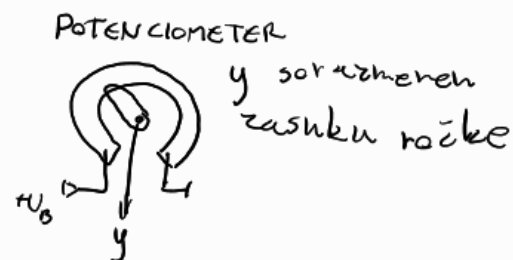
$$U_B R_2 = y (R_1 + R_2)$$

$y = U_B \frac{R_2}{R_1 + R_2}$

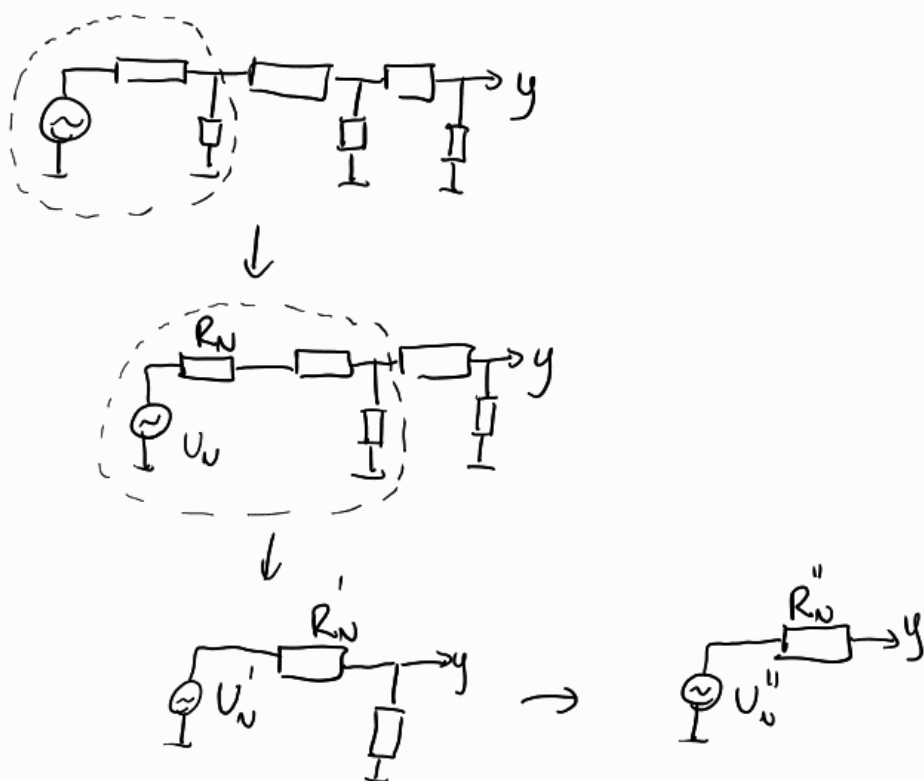
Delilnik napetosti - lahko dosežemo katerokoli napetost med 0 in U_B



$$y = U_B \frac{xR}{(1-x)R + xR} = x U_B$$



Theveninov teorem (nadomestno vezje)



→ Razumaljske simulacije

→ LINEARNA VEZJA

- definirajmo blok $x \rightarrow [A] \rightarrow z = A(x)$

(A je lahko kx , $\int$, $\frac{d}{dx}$, ...;
ne more biti x^2 , $\ln$, ...)
pri vходу z_x ni izhod z_z

- mehanizma $\rightarrow [A] \rightarrow [B] \rightarrow z = B(A(x))$

$\rightarrow [B] \rightarrow [A] \rightarrow z = A(B(x))$

- združitev $x \rightarrow [AB] \rightarrow z = AB(x)$

- seštevanje $\begin{matrix} x \rightarrow [A] \\ y \rightarrow [B] \end{matrix} \rightarrow z = Ax + By$

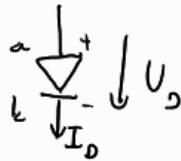
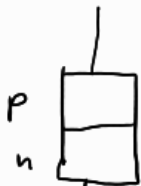
- povratna vezava $\begin{matrix} x \rightarrow \oplus \\ \oplus \rightarrow [A] \rightarrow y \\ y \rightarrow [B] \rightarrow \ominus \end{matrix}$
 $y = A(x - By)$
 $y + AB_y = Ax$

• če A ojačevalnik $\rightarrow \infty$:

$$y = x \frac{A}{1 + AB} = x \frac{1}{\frac{1}{A} + B}$$

$$\boxed{y = x \frac{1}{B}}$$

Dioda



$$I_D = I_{D0} (e^{U_D/U_T} - 1)$$

$$U_T = \frac{kT}{e} \text{ Termična temperatura}$$

$$U_T(\text{sobna } T) \approx 26 \text{ mV}$$

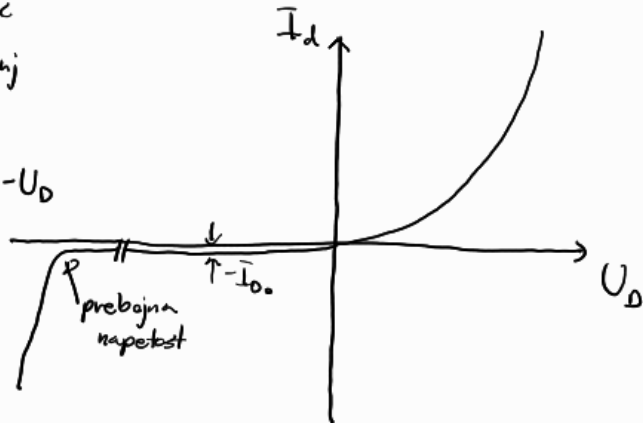
obogaten polprevodnik

- n ima elektrone z $1e^-$ v zun. lupini več

- p — " — — — — — manj

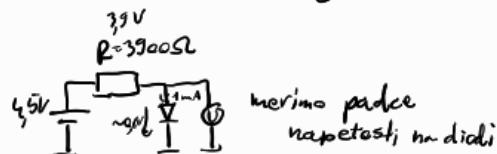
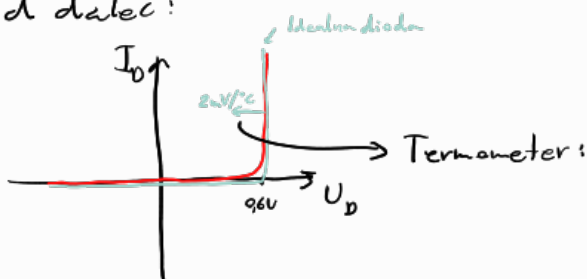
- lahko se jo uniži s prevelikim U_D in $-U_D$

• segrevanje $\rightarrow$ se stali



$\approx$ majhne I_D

od daleč:

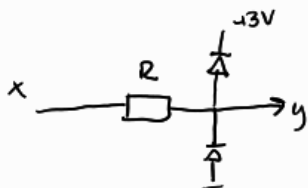


merimo padce napetosti na diodi

- Cellerjeva dioda $\frac{1}{A}$ - prebojna nap. je natanko določena, po abs. vr. je manjša od normalnih diod

- LED (Light emitting diode) $\frac{1}{A}$ - prag je višji (več od 1V, odvisno od barve)

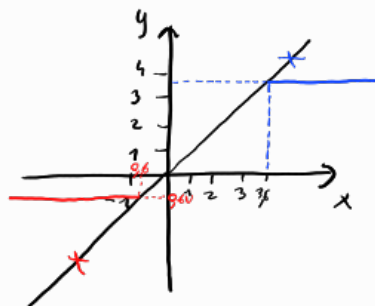
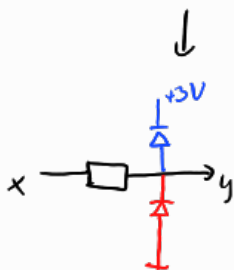
VEZJA Z DIODAMI



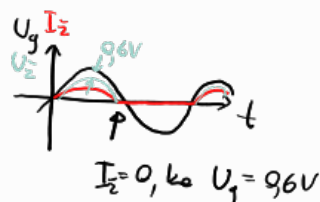
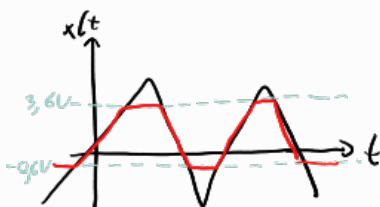
① preveri U_D

- $U_D > 0.6V$: dioda je vir nap. 0.6V

- $U_D < 0.6V$: računaj brez diode



$\Rightarrow$ omejili smo izhodno napetost med -0.6 in 3.6V

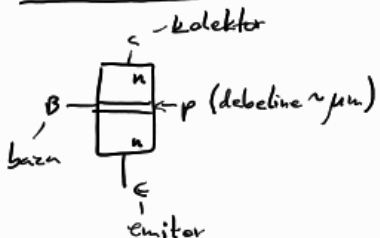




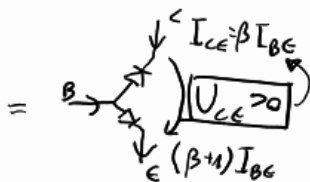
Gretov spoj / polnvalni usmernik



Tranzistor



V kolektorju je večji presežek e^- kot v emitorju



ker p sloj tanek → na diodo med B in C vpliva tudi tok v diodi B in E

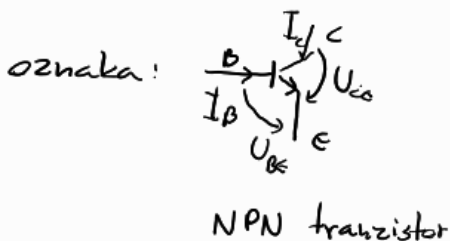
$$I_{BE} = I_{BE0} (e^{\frac{U_{BE}}{U_T}} - 1)$$

$$I_{CE} = \beta I_{BE}$$

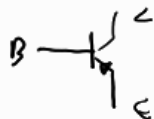
β -faktor ojačanja

$\beta \approx 100$ (tipično)

↑ velike tolerance (tudi 50%)



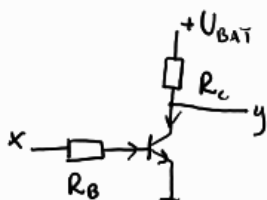
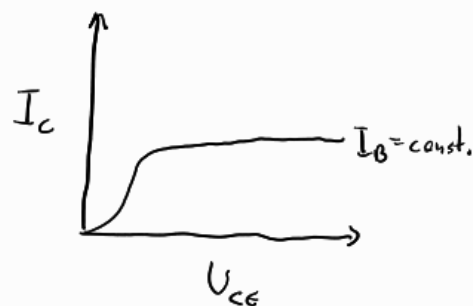
za PNP tranzistor velja enako, le predznaki so zamenjani



Uniženje:

- prevelika nap. skozi B
- prevelika negativna nap. iz C v B; iz E v B

Tok na C je neodv. od U_{CE}



1) če $x < 0$:

$$I_B = 0$$

$$I_C = 0 \Rightarrow y = U_{BAT}$$

2) če $x > 0$:

$I_B > 0$ ← točno je težko izračunati, se rešuje grafčno

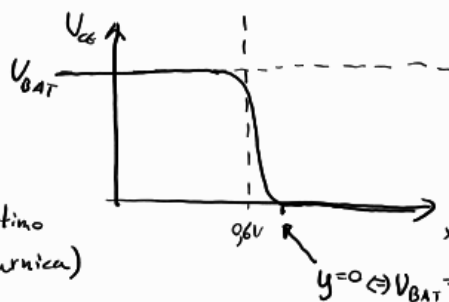


$$\Rightarrow U_{BE} = 0,6V$$

$$I_B = \frac{x - U_{BE}}{R_B} = I_C = \beta \frac{x - U_{BE}}{R_C}$$

$$\Downarrow$$

$$y = U_{BAT} - I_C R_C$$



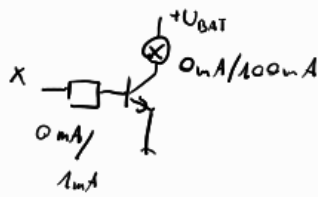
Uporaba tranzistorja:

1) Uporabi le $x > 0,6V$ in $x < 0,6V$

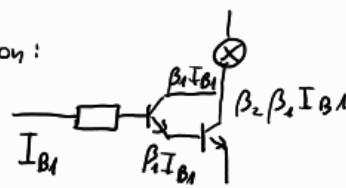
$$\text{npr. } x = \begin{cases} 0V \\ 5V \end{cases} \Rightarrow y = \begin{cases} U_{BAT} \\ 0V \end{cases} \Rightarrow \text{signal se obrne}$$

→ stikalo - zmajtni signalom x spustino večje tokove skozi R_C (npr. žarnica)

$$y = 0 \Leftrightarrow U_{BAT} = I_C R_C$$

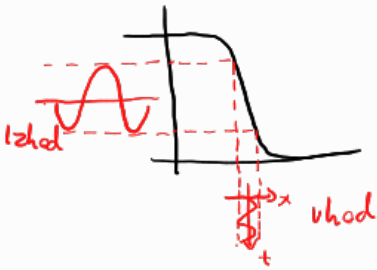


Darlington:



$$I_z = I_{B1} \beta_1 (\beta_2 + 1) \approx \underline{I_{B1} \beta_1 \beta_2}$$

2) ojačevalnik (če $x \approx 0,6V$)



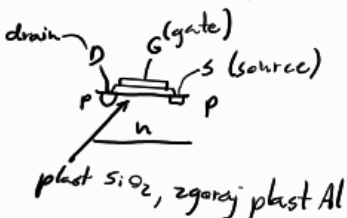
$$I_B = \frac{x - 0,6}{R_B}$$

$$y = U_{BAT} - \beta I_B R_C = U_{BAT} - \beta \frac{R_C}{R_B} (x - 0,6)$$

- ker je y odv. od $\beta \rightarrow$ ni uporabno, ker ima β velike tolerance
- ker je y odv. od $U_{BAT} \rightarrow$ ni uporabno, ker U_{BAT} odv. od polnosti baterije
- ker y odv. od $0,6V \rightarrow$ se spreminja glede na T ($2mV/^{\circ}C$)

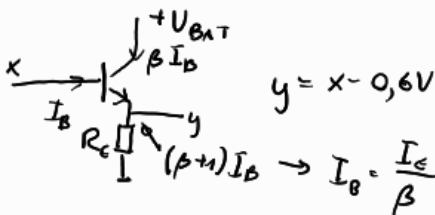
$\rightarrow$ je ok. npr. za ojačanje zvoka, ker lahko OT zanemarimo, ni pomembno, kako močno je ojačanje

MOST tranzistorji



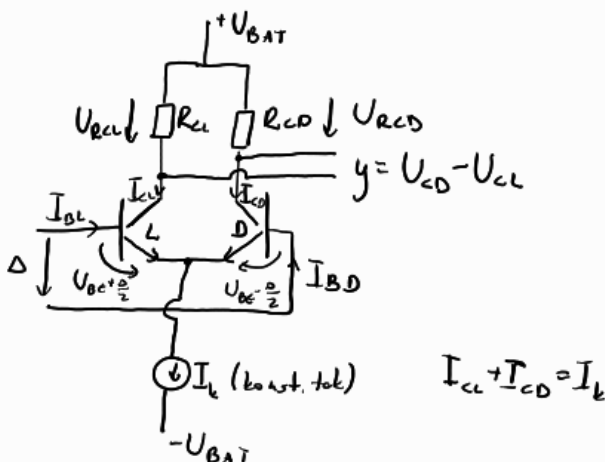
prednost, ker rabimo samo U_G , ne pa I_G

Emitorski sledilnik



$$y = x - 0,6V \quad I_B = \frac{I_E}{\beta} \text{ zmanjša breme predhodnega vezja}$$

Diferencialni par tranzistorjev kot ojačevalnik



$$\Delta = 0 \Rightarrow y = 0$$

$$\Delta > 0 \Rightarrow y < 0$$

$$\Delta < 0 \Rightarrow y > 0$$

$$y = (U_{BAT} - U_{RCD}) - (U_{BAT} - U_{RCL}) = I_{CL} R_{CL} - I_{CD} R_{CD} = \boxed{R_C (I_{CL} - I_{CD})}$$

$$R_{CL} = R_{CD}$$

$$I_{CL} + I_{CD} = I_k$$

$$I_{BL} = I_{BL0} (e^{U_{BE}/U_T} - 1) \rightarrow I_{CL} = \beta_L I_{BL0} e^{U_{BE}/U_T} \rightarrow \frac{I_{CL}}{I_{CD}} = \frac{\beta_L}{\beta_D} \cdot \frac{I_{BL0}}{I_{BD0}} \cdot \frac{e^{U_{BE}/U_T} \cdot e^{+U_T/2}}{e^{U_{BE}/U_T} \cdot e^{-U_T/2}}$$

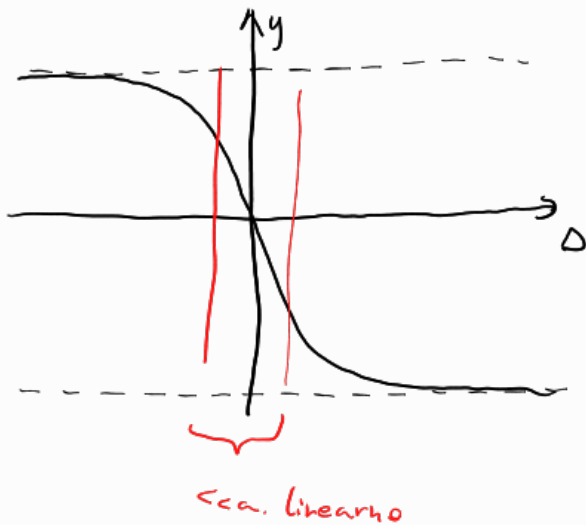
zan. ker sta kond. enaka

$$\rightarrow I_{CL} = I_{CD} e^{U_T/2}$$

$$\Rightarrow y = R_c \frac{e^{U_T/2} - 1}{e^{U_T/2} + 1} I_k = R_c I_k \tanh \frac{\Delta}{2U_T}$$

$$I_k = I_{CD} e^{U_T/2} + I_{CD}$$

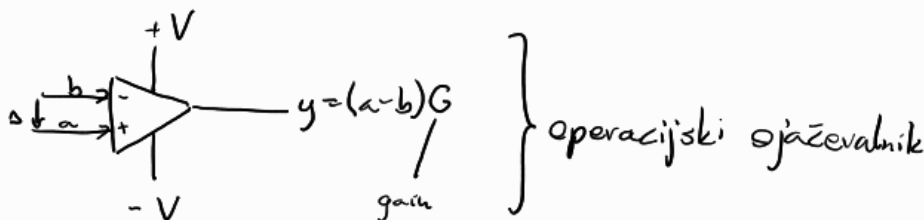
$$\rightarrow \begin{cases} I_{CD} = I_k \frac{1}{1 + e^{U_T/2}} \\ I_{CL} = I_k \frac{e^{U_T/2}}{1 + e^{U_T/2}} \end{cases}$$



Ojačevalnik ni odvisen od U_{BAT} , β in T

$\Rightarrow$ je boljši, ampak ojačen signal ni glede na GND, ampak le ΔU

Verzje dopolnimo, da gledamo ojačen signal glede na GND!



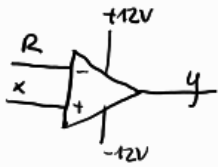
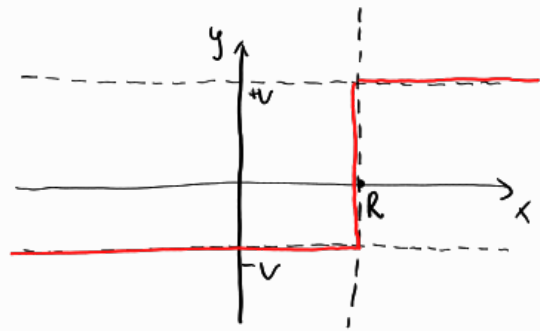
lastnost	realno	idealizirano
G	10^5	∞
vhodni tok I_b	$10^{-15} A - 10^{-4} A$	0
ΔV_h točen I_{off}	$0,1 \cdot I_b$	0
ΔV_h nap. U_{off}	$10^{-6} V - 10^{-2} V$	0
max. izh. nap. U_{sat}	$\pm (V - 1V)$	$\pm V$
max. nap. napor. t_V	$\pm 25V - \pm 50V$	—
max. izh. tok I_a	$\pm 20mA$	—
obseg vh.nap. a, b	V	—

$\rightarrow y = G(a - b - U_{off})$

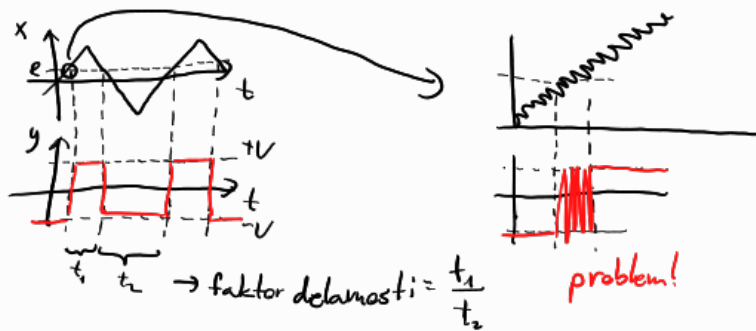
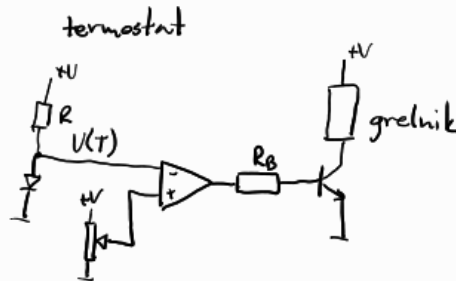
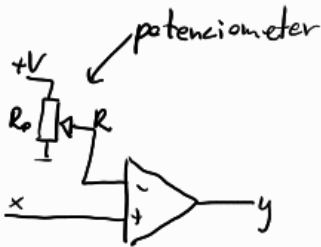
"bomo že našli pravi ojačevalnik"

Komparator

$$x \xrightarrow{R} \boxed{K} \rightarrow y = \begin{cases} +V & ; x > R \\ -V & ; x < R \end{cases}$$



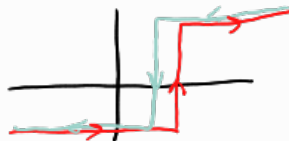
$$y = \infty (x - R) \quad (\text{največ } \pm V)$$



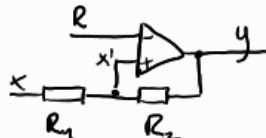
imamo



želim



POZITIVNA POUKATNA VEZAVA



$$R_2 \approx 100 R_1$$

$$I_{R_1} + I_{R_2} = 0$$

$$\frac{x - x'}{R_1} + \frac{y - x'}{R_2} = 0$$

$$x R_2 - x' R_2 + y R_1 - x' R_1 = 0$$

$$x' = \frac{R_2}{R_1 + R_2} x + \frac{R_1}{R_1 + R_2} y = R$$

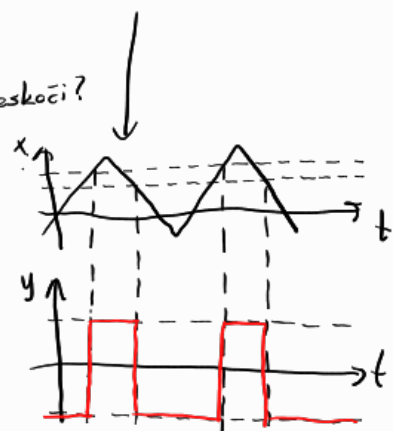
$$1) y = +V$$

$$x = R \left(\frac{R_1 + R_2}{R_2} \right) - V \frac{R_1}{R_2} = R \left(1 + \frac{1}{100} \right) - \frac{V}{100}$$

$$2) y = -V$$

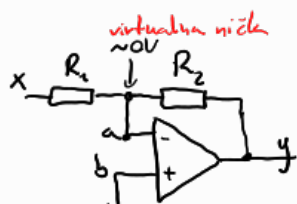
$$x = R \left(\frac{R_1 + R_2}{R_2} \right) + V \frac{R_1}{R_2} = R \left(1 + \frac{1}{100} \right) + \frac{V}{100}$$

$$\rightarrow \Delta = 2 \frac{V}{100}$$



NEG. POVRATNA VEZAVA

→ invertirani ojačevalnik (s stalno ojačitvijo)



$$I_{R_1} + I_{R_2} = 0$$

$$\frac{x-a}{R_1} + \frac{y-a}{R_2} = 0$$

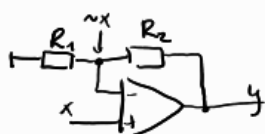
$$y = G(b-a) \rightarrow a = b - \frac{y}{G} = -\frac{y}{G} \rightarrow a = 0$$

$$\frac{x}{R_1} + \frac{y}{R_2} = 0$$

$$y = -x \frac{R_2}{R_1} \quad \text{do faktorja } \sim 100 \text{ dela super}$$

$$-\frac{R_2}{R_1} \text{ ojačanje}$$

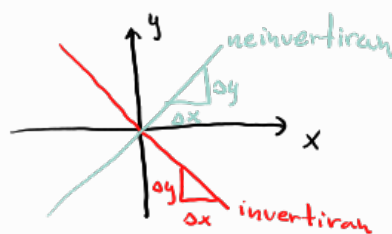
→ neinvertirani ojačevalnik



$$I_{R_1} + I_{R_2} = 0$$

$$\frac{x}{R_1} + \frac{x-y}{R_2} = 0$$

$$y = x \left(1 + \frac{R_2}{R_1}\right)$$



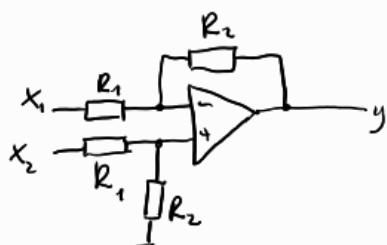
$$\text{ojačanje} = \frac{\Delta y}{\Delta x}$$

buffer



dobro, ker toka ne "krademo" toka z vhodne napetosti

Diferenčni ojačevalnik



$$a) x_2 = 0, x_1 \neq 0 \rightarrow y_1$$

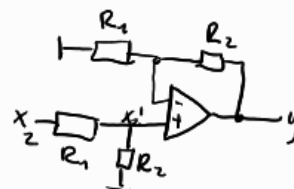
→ dobimo inv. ojač.

$$y_1 = -x \left(\frac{R_2}{R_1}\right)$$

$$c) y = y_1 + y_2$$

$$y = \frac{R_2}{R_1} (x_2 - x_1)$$

$$b) x_2 \neq 0, x_1 = 0 \rightarrow y_2$$

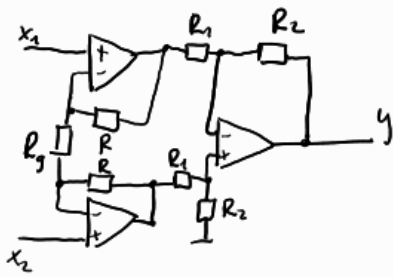


$$y_2 = x_2' \left(1 + \frac{R_2}{R_1}\right)$$

$$x_2' = x_2 \frac{R_2}{R_1 + R_2}$$

$$y_2 = x_2 \frac{R_2}{R_1 + R_2} \cdot \frac{R_1 + R_2}{R_1}$$

Instrumentacijski ojačevalnik



$$y = \frac{R_2}{R_1} \frac{R_1}{R} (x_2 - x_1)$$